

Real Estate CRM

Comprehensive Database Schema & REST API Architecture Specification

Project Name:	Real Estate CRM (EstateFlow)
GitHub Repository:	https://github.com/patturajan045/real-estate-crm
Deployment Platform:	Render Cloud Web Service (WSGI / Gunicorn)
Database Engine:	MongoDB 6.0+ (MongoEngine ODM & PyMongo Driver)
Authentication:	JWT (JSON Web Token) via Flask-JWT-Extended
Version:	2.0.0 Production Release

Executive Overview

This specification document provides an exhaustive, authoritative technical overview of the **Real Estate CRM** database design, document models, integrity constraints, indexing strategies, and complete REST API endpoint contracts. The architecture is engineered to guarantee strict concurrency safety (preventing race-condition double-bookings), complete role-based privilege isolation, and data consistency across high-volume real estate property management workflows.

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1. Database Architecture & Storage Model

The Real Estate CRM utilizes **MongoDB** as its primary document store, accessed through a high-performance dual-layer architecture: **MongoEngine 0.27.0** for structured Object-Document Mapping (ODM) and schema validation, and **PyMongo 4.6.2** for high-throughput aggregation and direct collection operations.

BaseDocument Pattern: All business models inherit from an abstract BaseDocument providing UUID-based string primary keys (avoiding opaque 12-byte hex BSON IDs in external APIs), ISO-8601 timestamp tracking (addedTime, updateTime), and automated recursive dictionary serialization via to_dict().

2. Data Dictionary: Collection Schemas

2.1 Collection: users (User Model)

Stores team accounts across the three defined privilege tiers with hashed credentials and status tracking.

Field	Type	Constraints	Default / Description
id	String	Primary Key, UUID4	Unique document identifier
name	String	Required, Max 120	Full Name / Login Username
email	Email	Required, Unique	Unique corporate email address
phoneNumber	String	Max 20	Contact phone number
password	String	Required	Werkzeug pbkdf2/scrypt hashed password
role	String	Super Admin Admin Sales Employee	Default: 'Sales Employee'
isActive	Boolean	Indexed	Default: True. False prevents login
addedTime	DateTime	Auto UTC	Account creation timestamp

2.2 Collection: projects (Project Model)

Top-level real estate developments and master masterplanned communities.

Field	Type	Constraints	Default / Description
id	String	Primary Key, UUID4	Unique project identifier
name	String	Required, Unique, Max 150	Project development name
description	String	Optional	Marketing and architectural overview
builder	String	Optional	Construction developer name
address, city, state	String	city Required	Geographical location details
status	String	Planning Under Construction Ready	Default: 'Under Construction'

2.3 Collection: buildings (Building Model)

Individual towers, wings, or structural blocks within a parent development.

Field	Type	Constraints	Default / Description
id	String	Primary Key, UUID4	Unique building identifier
project	Reference	Required, CASCADE delete	Parent Project foreign key

name	String	Required, Max 100	Tower / Wing name (e.g. 'Tower A')
totalFloors	Integer	Min 1	Total vertical floor levels
notes	String	Optional	Tower specific amenities / notes

2.4 Collection: units (Unit Model)

Individual marketable residential/commercial units. Subject to atomic state changes.

Field	Type	Constraints	Default / Description
id	String	Primary Key, UUID4	Unique unit identifier
project	Reference	Required, CASCADE delete	Foreign reference to Project
building	Reference	Required, CASCADE delete	Foreign reference to Building
unitNumber	String	Required, Max 50	Unit label (e.g. 'A-101', 'PH-12')
floor	Integer	Required	Floor number
unitType	String	1BHK-4BHK Penthouse Villa Commercial	Default: '2BHK'
carpetAreaSqFt	Float	Required	Net usable carpet area in sq ft
price	Float	Required, Min 0.0	Base listing price in USD (\$)
status	String	Available Blocked Booked Sold	Default: 'Available'. Mutated atomically

2.5 Collection: leads (Lead & LeadNote Models)

Customer prospects with stage progression, financial budget, and embedded interaction notes.

Field	Type	Constraints	Default / Description
id	String	Primary Key, UUID4	Unique lead identifier
customerName	String	Required, Max 120	Customer full name
email, phoneNumber	String	Required	Direct contact coordinates
stage	String	7 Defined Pipeline Stages	Default: 'New'. Indexed
budgetMin / Max	Float	Optional	Buyer pricing range
assignedTo	Reference	NULLIFY delete rule	Assigned sales representative User ID
nextFollowUpDate	DateTime	Indexed	Scheduled follow-up date and time
notes	List[Embedded]	LeadNote EmbeddedDocument	Timeline: authorId, authorName, content, time

2.6 Collection: bookings (Booking Model)

Formal property reservation contracts connecting a customer lead with an inventory unit.

Field	Type	Constraints	Default / Description
id	String	Primary Key, UUID4	Unique booking record identifier
bookingReference	String	Unique, Indexed	E.g. 'BK-20260908-A9C3F1'
lead	Reference	Required, CASCADE	Purchasing customer Lead reference
unit	Reference	Required, Partial Unique	Reserved Unit reference
bookedBy	Reference	NULLIFY delete rule	Executing Sales Employee User ID
agreementValue	Float	Required	Final negotiated agreement price
bookingAmount	Float	Default 0.0	Token / advance deposit received
status	String	Confirmed Cancelled	Default: 'Confirmed'

2.7 Collection: page_contents (Dynamic CMS Model)

Powers no-code dynamic branding, headings, subheadings, and navigation labels editable exclusively by Super Admin.

Field	Type	Constraints	Default / Description
page	String	Required	Page module key (e.g. 'dashboard', 'login')
sectionKey	String	Required, Unique	Unique binding identifier (e.g. 'leads_title')
label	String	Required	Human-readable label shown in settings UI
content	String	Required	Current live rendered text / title / paragraph
contentType	String	title subtitle heading paragraph	Semantic content category

2.8 Collection: notifications (In-App Alerts Model)

Real-time follow-up and booking assignment notifications delivered to sales staff and admins.

Field	Type	Constraints	Default / Description
recipient	Reference	Required, CASCADE	Target User recipient reference
sender	Reference	NULLIFY delete rule	Assigning Manager User reference
senderName	String	Default: 'Administrator'	Snapshot of assigner display name
title, message	String	Required	E.g. 'Follow-up assigned by System Admin'
entityType, entityId	String	lead booking	Direct deep-linking target entity
isRead	Boolean	Default: False	Unread tracking flag for topbar badge

3. Concurrency Protection & Indexing Strategy

Atomic Concurrency Guard: To prevent simultaneous double-booking of property units by concurrent agents, the booking engine executes an atomic check-and-set query directly at the database level:

```
updated_count = Unit.objects(id=unit_id,
status=Unit.STATUS_AVAILABLE).update_one(set__status=Unit.STATUS_BOOKED)
if updated_count == 0:
    return jsonify({'status': 'error', 'message': 'Double booking prevented! Unit already
booked.'}), 409
```

Unique & Compound Indexes:

- Booking.unit: Enforces a partial unique index where status == 'Confirmed'.
- Unit: Compound index on ('building', 'unitNumber') ensures unit numbers are unique within towers.
- Building: Compound index on ('project', 'name').
- Lead: Compound query indexes on ['stage', 'assignedTo', 'nextFollowUpDate', '-addedTime'].

4. REST API Specification & Endpoint Contracts

All endpoints return JSON responses adhering to the standard envelope: `{'status': 'success'|'error', 'data': {...}, 'message': '...'}`. Endpoints marked with ■ require an HTTP Authorization: Bearer <JWT> header.

4.1 Authentication & User Management Endpoints

Method & Endpoint	Auth / Role	Payload / Params	Success / Status
POST /api/auth/register	Public	{name, email, password, phoneNumber}	201 Created (JWT Token + User)
POST /api/auth/login	Public	{email/username, password}	200 OK (JWT Token + User)
GET /api/auth/me	■ Any Role	None	200 OK (Current Profile)
POST /api/auth/logout	■ Any Role	None	200 OK (Session Cleared)
GET /api/users/	Public / Any	None	200 OK (User Directory Array)
POST /api/users/	■ Admin / Super	{name, email, password, role, phone}	201 Created (User Object)
PUT /api/users/<id>	■ Admin / Super	{name, email, role, phone, password}	200 OK (Updated User)
DELETE /api/users/<id>	■ Admin / Super	?permanent=true (optional)	200 OK (Deactivates / Deletes)

4.2 Lead Pipeline & Activity Endpoints

Method & Endpoint	Auth / Role	Payload / Params	Success / Status
GET /api/leads/	Public / Any	?stage=New&search;=John&assignedTo;=...	200 OK (Filtered Leads List)
POST /api/leads/	■ Optional JWT	{customerName, email, phoneNumber, assignedTo, stage, budgetMin, budgetMax, nextFollowUpDate}	201 Created (Triggers in-app assignment notification)
GET /api/leads/<id>	Public / Any	None	200 OK (Lead with Notes)
PUT /api/leads/<id>	■ Optional JWT	{customerName, stage, assignedTo, ...}	200 OK (Triggers reassignment alert)
DELETE /api/leads/<id>	Public / Any	None	200 OK (Lead Removed)
POST /api/leads/<id>/notes	■ Optional JWT	{authorId, authorName, content, stage, nextFollowUpDate}	201 Created (Appends Note to Lead Timeline)

4.3 Inventory Management & Atomic Bookings

Method & Endpoint	Auth / Role	Payload / Params	Success / Status
GET /api/projects/	Public / Any	None	200 OK (All Master Projects)
POST /api/projects/	Public / Any	{name, city, address, status, ...}	201 Created (Project Object)
GET /api/buildings/	Public / Any	?project_id=<id>	200 OK (Building Towers List)
POST /api/buildings/	Public / Any	{project, name, totalFloors, notes}	201 Created (Building Object)
GET /api/units/	Public / Any	?project_id=&status;=Available&unit_type=...	200 OK (Filtered Units List)
POST /api/units/	Public / Any	{project, building, unitNumber, floor, unitType, carpetAreaSqFt, price}	201 Created (Unit Object)
POST /api/bookings/	■ Optional JWT	{lead, unit, bookedBy, agreementValue, bookingAmount, paymentMethod}	201 Created (Locks Unit to Booked) 409 Conflict if race condition
GET /api/bookings/	Public / Any	?status=Confirmed	200 OK (Bookings List)
POST /api/bookings/<id>/cancel	Public / Any	{cancellationReason}	200 OK (Unit status → Available)

4.4 Analytics, Dynamic CMS & In-App Notifications

Method & Endpoint	Auth / Role	Payload / Params	Success / Status
GET /api/dashboard/stats	Public / Any	None	200 OK (Leads, Units, Bookings, Revenue, Urgent Follow-ups)
GET /api/dashboard/pipeline	Public / Any	None	200 OK (Counts by 7 stages)
GET /api/dashboard/follow-ups	Public / Any	None	200 OK (Urgent/Overdue tasks)
GET /api/cms/content	Public	None	200 OK (All 48 UI Text Keys)
PUT /api/cms/content/batch	■ Super Admin Only	{updates: {key: value, ...}}	200 OK 403 Forbidden for Admin/Sales
POST /api/cms/reset	■ Super Admin Only	None	200 OK (Restores Defaults) 403 Forbidden for others
GET /api/notifications/	■ Any Auth User	None	200 OK (unreadCount + Array)
POST /api/notifications/<id>/read	■ Any Auth User	None	200 OK (Item marked as read)
POST /api/notifications/mark-all-read	■ Any Auth User	None	200 OK (All unread cleared)
GET /api/health	Public	None	200 OK (Health operational check)

5. Security, JWT Lifecycle & Error Standards

Security Standard: All authenticated requests pass a signed JWT Bearer token in the Authorization header. Tokens carry the user's primary key identity. Password hashes are generated using Werkzeug's secure hashing algorithm and are completely stripped from API responses. In-memory session expiry triggers seamless client redirection to /login.

HTTP Error Codes:

- **200 / 201:** Successful query execution or entity creation.
- **400 Bad Request:** Missing required input fields or invalid payloads.
- **401 Unauthorized:** Missing, expired, or invalid JWT authentication token.
- **403 Forbidden:** Role restriction violation (e.g. non-Super Admin attempting CMS modifications).
- **404 Not Found:** Referenced entity or route does not exist.
- **409 Conflict:** Unique constraint collision (e.g. duplicate email, duplicate unit number, or concurrent double booking).
- **500 Server Error:** Unhandled system exception with sanitized user-facing error message.